

6 Botensilimab (Fc-enhanced anti-cytotoxic lymphocyte-association protein-4 antibody) Plus Balstilimab (anti-PD-1 antibody) in Patients With Relapsed/Refractory Metastatic Sarcomas

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ABSTRACT

PURPOSE Outcomes for patients with advanced sarcomas are poor and there is a high unmet need to develop novel therapies. The purpose of this phase I study was to define the safety and efficacy of botensilimab (BOT), an Fc-enhanced anti-cytotoxic lymphocyte-association protein-4 antibody, plus balstilimab (BAL), an anti-PD-1 antibody, in advanced sarcomas.

METHODS BOT was administered intravenously (IV) at 1 mg/kg or 2 mg/kg once every 6 weeks in combination with BAL IV at 3 mg/kg once every 2 weeks for up to 2 years. The primary end point was to determine dose-limiting toxicities during the dose-escalation period. Secondary end points include objective response rate (ORR), duration of response (DOR), disease control rate, and progression-free survival (PFS) by RECIST 1.1. Exploratory end points include assessing patient biomarkers including tumor mutational burden, cytokines, and PD-L1 expression.

RESULTS Overall, 64 patients with sarcoma were treated; all were evaluable for safety and 52 for efficacy. The most common treatment-related adverse event (TRAE) was diarrhea/colitis occurring in 35.9% of patients, with grade 3 in 6.3% of patients. No grade 4 or 5 TRAEs were reported. For all evaluable patients, ORR was 19.2% (95% CI, 9.6 to 32.5), and 27.8% (95% CI, 9.7 to 53.5) for evaluable patients with angiosarcoma (n = 18); 33.3% in visceral and 22.2% in cutaneous subtypes. Median PFS for evaluable patients was 4.4 months (95% CI, 2.8 to 6.1), with a 6-month PFS rate of 36% (95% CI, 22 to 50) and a median DOR of 21.7 months (95% CI, 1.9 to not reached).

CONCLUSION The combination of BOT/BAL demonstrated promising efficacy and safety in a large cohort of heavily pretreated sarcoma patients. This encouraging activity warrants further investigation (ClinicalTrials.gov identifier: [NCT03860272](https://clinicaltrials.gov/ct2/show/study/NCT03860272)).

ACCOMPANYING CONTENT

 Listen to the podcast by Dr Westin and Dr Wilky at <https://ascopubs.org/doi/botensilimab-plus-balstilimab-advanced-sarcomas>

 Appendix

 Data Sharing Statement

 Data Supplement

 Protocol

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INTRODUCTION

Sarcomas are heterogeneous neoplasms, with more than 150 different subtypes of bone and soft tissue tumors that collectively make up approximately 1% of all adult cancers.¹ In the metastatic setting, overall survival (OS) with treatment remains poor, with 5-year survival rates approximately 16%, thus novel treatments are urgently needed, including immunotherapies that have the potential for durable responses but are not approved for most subtypes.² Common adult sarcomas including leiomyosarcoma (LMS) and dedifferentiated liposarcoma (dLPS)

have response rates <10% to PD-1 monotherapy,^{3,4} likely because of poor tumor immunogenicity, with low tumor mutational burden (TMB), suppressive microenvironments, dysregulated antigen presentation machinery, and low expression of immune-related genes.⁵ An exception is cutaneous angiosarcoma (cAS) arising in the face and scalp, which exhibits an ultraviolet (UV) genetic signature akin to melanoma.⁶ In a small study of ipilimumab and nivolumab, tumor responses were noted in four of eight patients with cAS, including three of five patients with face/scalp disease.⁷ However, only one of seven patients responded with noncutaneous or visceral angiosarcomas

CONTEXT

Key Objective

To define the safety and efficacy of botensilimab (BOT) plus balstilimab (BAL) in advanced sarcomas.

Knowledge Generated

Treatment with an Fc-enhanced anti-cytotoxic lymphocyte-association protein-4 antibody BOT plus BAL (anti-PD-1) induced durable responses across immunologically cold soft tissue sarcoma types, including visceral angiosarcoma and leiomyosarcoma, historically unlikely to respond to traditional checkpoint blockade. Diarrhea/colitis was the most common immune-mediated treatment-related adverse event, manageable with the use of steroids and anti-tumor necrosis factor alpha therapy.

Relevance (R.G. Maki)

There are no regulator-approved immune checkpoint inhibitors for patients with metastatic sarcoma beyond alveolar soft part sarcoma in the United States. This study provides a potential route forward for use of immunotherapy in soft tissue sarcomas, most of which have low tumor mutational burden and are PD-L1 –negative.*

*Relevance section written by JCO Associate Editor Robert G. Maki, MD, PhD, FACP, FASCO.

(vAS), which are driven by alternative genetic drivers and are considered immunologically cold.

Botensilimab (BOT), a multifunctional anti-cytotoxic lymphocyte-association protein (CTLA-4) antibody, retains an Fc-enhanced IgG1-region producing superior binding to activating Fc-gamma receptors (FcGRs) compared with conventional anti-CTLA-4 antibodies such as ipilimumab.⁸ This promotes superior engagement of T cells with antigen-presenting cells (APCs) and natural killer cells, resulting in enhanced T-cell priming, activation, and memory formation.⁸ Balstilimab (BAL) is a fully humanized anti-PD-1 antibody with activity comparable with approved PD-1 inhibitors.^{9,10} Reported results from a phase I, open-label study of BOT ± BAL (ClinicalTrials.gov identifier: [NCT03860272](https://clinicaltrials.gov/ct2/show/study/NCT03860272)) have shown remarkable outcomes in a variety of immune-cold tumors unlikely to respond to traditional PD-1/CTLA-4 inhibitors, such as microsatellite stable metastatic colorectal cancer (MSS mCRC), and tumors refractory to previous checkpoint blockade.^{11,12}

Here, we report promising safety and efficacy results of BOT/BAL for patients with advanced/metastatic sarcomas treated in a large phase I trial. This combination has the potential to become a new treatment for this group of largely immunologically cold sarcomas, requiring confirmatory studies.

METHODS

Study Design and Participants

C-800-01 is an open-label, phase Ib, multicenter study evaluating the safety, tolerability, efficacy, and pharmacokinetic/pharmacodynamic profiles of BOT ± BAL (ClinicalTrials.gov identifier: [NCT03860272](https://clinicaltrials.gov/ct2/show/study/NCT03860272), see Protocol, Amendment 8).

Patients were enrolled and treated between October 1, 2020, and January 24, 2024, at 12 sites across the United States and the United Kingdom. Patients with sarcoma were enrolled to the dose-escalation portion of the study, which was a 3 + 3 design, and within a later expansion cohort. The full study design, results of dose escalation, and efficacy and safety results for the largest tumor cohort (MSS mCRC) have been recently reported.¹²

Eligible patients were age 18 years and older with measurable disease per RECIST version 1.1 and had an Eastern Cooperative Oncology Group (ECOG) performance status (PS) of 0 or 1 with adequate end-organ function. Previous immune checkpoint inhibitors (ICI) were permitted. Patients must have had a confirmed diagnosis of refractory sarcoma for which no standard therapy was available or where standard therapies previously failed. The study was conducted in compliance with the Declaration of Helsinki and International Conference on Harmonisation Guidelines for Good Clinical Practice and was approved by the institutional review board at each institution. Patients provided informed written consent.

Procedures

During dose escalation, BOT was administered intravenously (IV) once every 3 weeks or once every 6 weeks at dose levels of 0.1–3 mg/kg for up to 2 years. BAL was administered IV once every 2 weeks (up to 2 years) at a dose level of 3 mg/kg when combined with BOT. Patients enrolled on the dose-expansion portion of the study received 1 or 2 mg/kg of BOT (once every 6 weeks) up to 2 years in combination with BAL 3 mg/kg (once every 2 weeks) up to 2 years. Patients were permitted to crossover from BOT monotherapy to the combination including treatment beyond progression.

End Points

The primary end point was safety and toxicity, by determining the occurrence of dose-limiting toxicities (DLTs) during dose escalation as previously reported. Assessment of severity and duration of treatment-emergent adverse events (TEAEs) and laboratory abnormalities, according to the National Cancer Institute's Common Terminology Criteria for Adverse Events (NCI-CTCAE version 5.0) was a secondary end point.

Additional secondary end points included (1) objective response rate (ORR); (2) duration of response (DOR); (3) disease control rate (DCR), defined as a complete response (CR) or partial response (PR) or stable disease (SD) of at least 6 weeks; and (4) progression-free survival (PFS), all per RECIST 1.1 on the basis of investigator assessment. The clinical benefit rate (CBR) was also reported and was defined as a best response of CR, PR, or SD for ≥ 24 weeks. Imaging was performed once every 6 weeks (± 3 days) from the first treatment. To better describe patterns of response with immunotherapy including pseudoprogression, responses by modified RECIST 1.1 for immune-based therapeutics (iRECIST; specifically designed for assessing response to immunologic agents v chemotherapeutics) on the basis of investigator assessment were also collected as part of an exploratory analysis.

Statistical Analysis

The data cutoff date was July 25, 2024, and the safety analysis population includes all patients who received at least one dose of study drug ($n = 64$). Safety end points were tabulated by dose level, using descriptive statistics, and the severity of adverse events was graded using NCI-CTCAE whenever possible. The efficacy evaluable set includes all treated patients with at least one postbaseline 6-week tumor imaging scan ($n = 52$). Descriptive statistics were used in general to summarize trial results, including means, medians, ranges, and measures of variability. Qualitative variables were summarized by counts and percentages. Best overall response was defined as a confirmed CR or PR. For patients with a CR or PR, DOR was summarized using Kaplan-Meier methods, as were PFS and OS for all patients. Median time to event(s) and 95% CI for the median were provided.

Correlative Studies

TMB and FcGR Genotyping

TMB was determined through local clinical testing at designated reference laboratories. All TMB assessments were conducted using validated next-generation sequencing panels, with results reported as mutations/megabase (mut/Mb). FcGR genotyping was performed as described.⁸ Genomic DNA was isolated from blood using QIAamp Midi Kit and subjected to TaqMan Genotyping to

detect single-nucleotide polymorphism (SNP) rs396991 (FCGR3A). Data acquisition was performed using ViiA7 or QuantStudio instruments.

Cytokine Analysis

Pretreatment plasma was prepared from blood collected in NaHep tubes and processed immediately at clinical sites. Plasma aliquots were stored at -80°C . Interleukin-6 (IL-6) levels in plasma were measured using the Meso Scale Discovery V-PLEX Plus Proinflammatory Panel 1 electrochemiluminescence assay (K15049G; Precision for Medicine, Frederick, MD).

Tumor PD-L1 Immunohistochemistry

PD-L1 protein expression was evaluated in pretreatment biopsy specimens using immunohistochemistry 28-8 pharmDx qualitative immunohistochemical assay (Dako/Agilent, Cat#SK005) and examined by a pathologist. Tissue samples with <100 tumor cells/slide were considered nonevaluable. PD-L1 expression in all evaluable samples was assessed by combined proportional score (CPS) as described.⁸

RESULTS

Dose Escalation and Dose Expansion

During the dose-escalation portion of the study (April 1, 2019–August 31, 2023; data cut-off, July 25, 2024), 83 patients were treated with BOT monotherapy or combination BOT/BAL, including six patients with sarcoma. Four received BOT monotherapy, including two who initially received BOT monotherapy IV at 2 mg/kg once every 3 weeks and crossed over to the combination of BOT IV at 2 mg/kg once every 6 weeks plus BAL IV at 3 mg/kg once every 2 weeks; and two received the 1 or 2 mg/kg BOT/BAL combination (BOT IV at 1 or 2 mg/kg once every 6 weeks and BAL IV at 3 mg/kg once every 2 weeks). No DLTs occurred during the dose-escalation period and a maximum tolerated dose was not reached (NR).¹² The recommended phase II dose is expected to be either the 1 or 2 mg/kg dose of BOT for sarcoma.

During dose expansion, 63 patients with sarcoma were treated, with 60 receiving 1 or 2 mg/kg BOT plus BAL (BOT IV at 1 or 2 mg/kg once every 6 weeks and BAL IV at 3 mg/kg once every 2 weeks), and three who received a fixed dose of BOT IV at 150 mg once every 6 weeks and BAL IV at 450 mg once every 3 weeks, but who were not included in the analysis. Overall, 64 patients were treated and evaluable for safety and 52 patients had at least one postbaseline 6-week imaging scan and were efficacy-evaluable (Fig 1).

Patients

Demographics of the safety analysis population ($n = 64$) were consistent with general trends among patients with

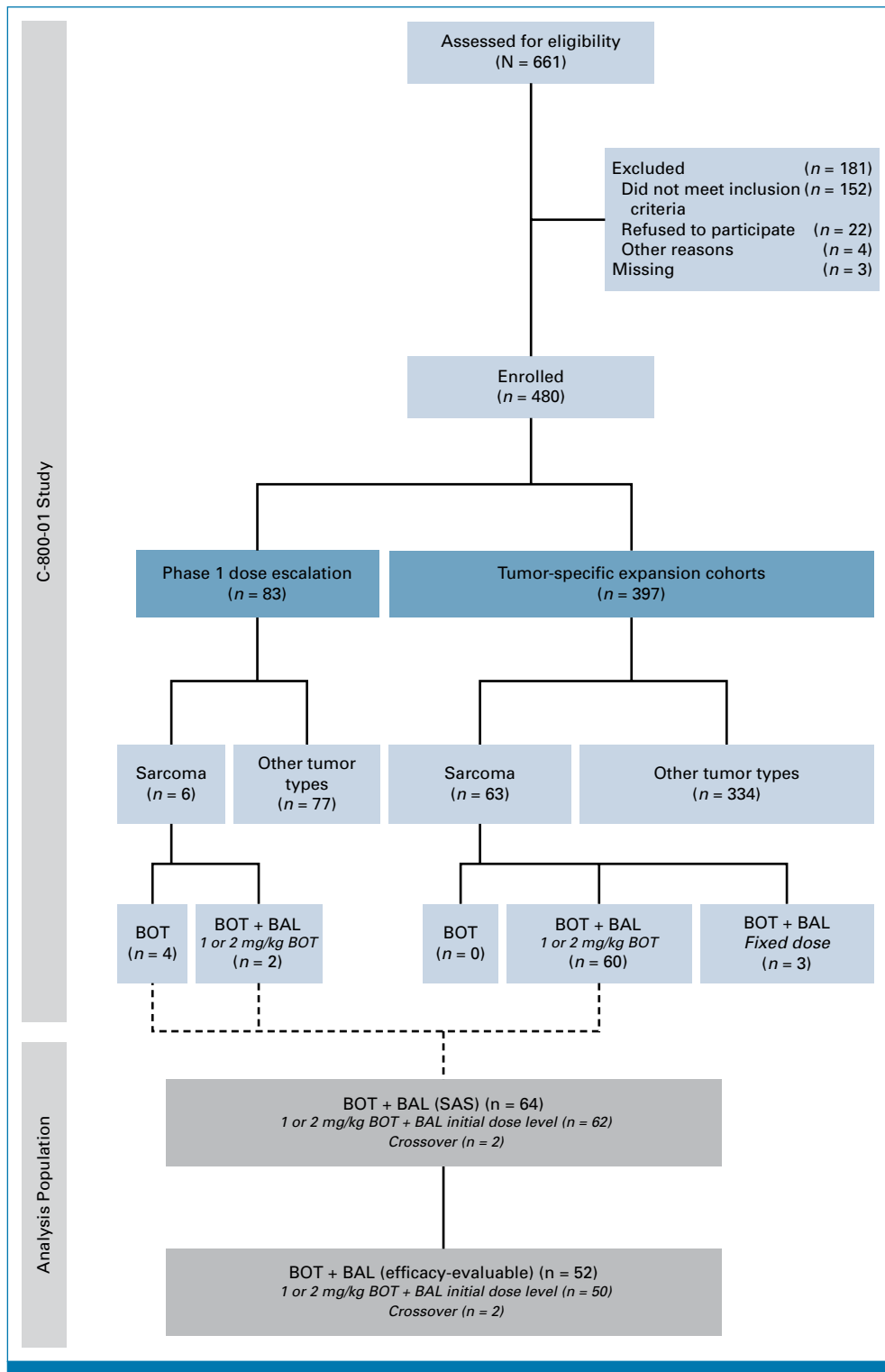


FIG 1. Flow diagram of the C-800-01 study and all treated patients with relapsed/refractory sarcomas ($n = 64$). BAL, balstilimab; BOT, botensilimab; SAS, safety analysis population.

sarcoma (Table 1): the median age was 61 years (range, 30–81), most patients were female (65.6%), with an ECOG PS of 0 (40.6%) or 1 (59.4%). Patients were heavily pretreated, with a median of three lines of previous therapy (range, 0–10). Fifteen percent had received previous PD-(L)1-directed therapy, and 3.3% had received previous CTLA-4

blockade, with one patient previously treated with both. The most common sarcoma subtype was angiosarcoma (AS; $n = 25$), including 13 patients with cAS (20.3%) and 12 with vAS (18.8%). LMS was the next most common ($n = 22$), with 16 patients with soft tissue LMS (25.0%) and six with uterine LMS (9.4%). Additional subtypes included high-grade

TABLE 1. Demographics and Baseline Characteristics in All Treated Patients With Sarcoma (n = 64)

Patient Characteristic	1 mg/kg Combo (n = 47)	2 mg/kg Combo (n = 15)	Crossover (n = 2)	Overall (n = 64)
Age, years, median (range)	60.0 (30.0-81.0)	69.0 (31-80.0)	51.00 (49.0-53.0)	61.00 (30.0-81.0)
Sex, No. (%)				
Female	34 (72.3)	7 (46.7)	1 (50.0)	42 (65.6)
Male	13 (27.7)	8 (53.3)	1 (50.0)	22 (34.4)
ECOG PS, No. (%)				
0	18 (38.3)	6 (40.0)	2 (100.0)	26 (40.6)
1	29 (61.7)	9 (60.0)	0	38 (59.4)
Previous lines of therapy				
Median (range)	3 (0-10)	2 (0-8)	3.5 (2-5)	3 (0-10)
0, No. (%)	3 (6.4)	1 (6.7)	0	4 (6.3)
1, No. (%)	8 (17.0)	4 (26.7)	0	12 (18.8)
2, No. (%)	10 (21.3)	3 (20.0)	1 (50.0)	14 (21.9)
≥3, No. (%)	26 (55.3)	7 (46.7)	1 (50.0)	34 (53.1)
Previous therapies, No. (%)				
Previous PD-(L)1 or CTLA-4 therapy	4 (8.5)	6 (40.0)	0	10 (15.6)
Yes	40 (85.1)	8 (53.3)	2 (100.0)	50 (78.1)
No	3 (6.4)	1 (6.7)	0	4 (6.3)
Missing				
Radiotherapy				
Yes	30 (63.8)	9 (60.0)	1 (50.0)	40 (62.5)
No	17 (36.2)	6 (40.0)	1 (50.0)	24 (37.5)
Sarcoma subtype, No. (%)				
cAS	4 (8.5)	9 (60.0)	0	13 (20.3)
vAS	8 (17.0)	4 (26.7)	0	12 (18.8)
dLPS	5 (10.6)	0	1 (50.0)	6 (9.4)
High-grade UPS	6 (12.8)	2 (13.3)	0	8 (12.5)
Soft tissue LMS	15 (31.9)	0	1 (50.0)	16 (25.0)
Uterine LMS	6 (12.8)	0	0	6 (9.4)
Other	3 (6.4)	0	0	3 (4.7)

Abbreviations: cAS, cutaneous angiosarcoma; CTLA-4, cytotoxic lymphocyte-association protein-4; dLPS, dedifferentiated liposarcoma; ECOG PS, Eastern Cooperative Oncology Group performance status; LMS, leiomyosarcoma; PS, performance status; UPS, undifferentiated pleomorphic sarcoma; vAS, visceral angiosarcoma.

undifferentiated sarcomas (n = 8; 12.5%), undifferentiated pleomorphic sarcoma (UPS; n = 4; 6.3%), dLPS (n = 6; 9.4%), and other, which included one patient with myxoid liposarcoma, one with osteosarcoma/spindle cell sarcoma, and one with synovial sarcoma (4.7%).

Safety

Safety is reported for all 64 patients who received at least one dose of treatment (Table 2; Appendix Table A1, online only). Forty-seven patients received BOT IV at 1 mg/kg once every 6 weeks plus BAL IV at 3 mg/kg once every 2 weeks, and 15 received 2 mg/kg BOT IV once every 6 weeks plus BAL IV at 3 mg/kg once every 2 weeks. Two patients received BOT monotherapy IV at 2 mg/kg once every 3 weeks and crossed over to 2 mg/kg BOT/BAL (BOT IV at 2 mg/kg once every 6 weeks plus BAL IV at 3 mg/kg once every 2 weeks). All patients experienced at least one TEAE (62.5% grade 3;

Appendix Table A1) and 82.8% experienced at least one treatment-related adverse event (TRAE; 17.2% grade 3; Table 2). No grade 4 or 5 TRAEs were reported. The most common TRAE was diarrhea/colitis occurring in 35.9% of patients (grade 3, 6.3%). Incidence was similar among patients receiving 1 or 2 mg/kg of BOT. Other common TRAEs occurring in more than 10.0% of patients included fatigue (any grade 26.6%, 12.5% grade 3), pyrexia (any grade 21.9%, no grade 3), chills (any grade 17.2%, no grade 3), rash (any grade 17.2%, 1.6% grade 3), rash maculopapular (any grade 14.1%, no grade 3), myalgia (any grade 10.9%, 1.6% grade 3), and hypothyroidism (any grade 10.9%; no grade 3). There were no TRAEs of pneumonitis, myocarditis, or hypophyinitis reported in this cohort.

Approximately 51.6% of patients experienced at least one immune-mediated TRAE, defined as related to treatment and subsequently treated with steroids (of any dose) or other

TABLE 2. TRAEs in At Least 10% of All Treated Patients With Sarcoma (n = 64)

Adverse Event	1 mg/kg Combo (n = 47), No. (%)		2 mg/kg Combo (n = 15), No. (%)		Crossover (n = 2), No. (%)		Overall (n = 64), No. (%)	
	Any Grade	Grade 3	Any Grade	Grade 3	Any Grade	Grade 3	Any Grade	Grade 3
Any TRAE	37 (78.7)	6 (12.8)	14 (93.3)	5 (33.3)	2 (100.0)	0	53 (82.8)	11 (17.2)
Diarrhea/colitis	17 (36.2)	3 (6.4)	5 (33.3)	1 (6.7)	1 (50.0)	0	23 (35.9)	4 (6.3)
Fatigue	11 (23.4)	0	5 (33.3)	1 (6.7)	1 (50.0)	0	17 (26.6)	1 (1.6)
Pyrexia	11 (23.4)	0	3 (20.0)	0	0	0	14 (21.9)	0
Chills	6 (12.8)	0	4 (26.7)	0	1 (50.0)	0	11 (17.2)	0
Rash	7 (14.9)	0	4 (26.7)	1 (6.7)	0	0	11 (17.2)	1 (1.6)
Rash maculopapular	7 (14.9)	0	2 (13.3)	0	0	0	9 (14.1)	0
Immune-mediated enterocolitis	5 (10.6)	1 (2.1)	3 (20.0)	1 (6.7)	0	0	8 (12.5)	2 (3.1)
Nausea	4 (8.5)	0	4 (26.7)	1 (6.7)	0	0	8 (12.5)	1 (1.6)
Hypothyroidism	4 (8.5)	0	3 (20.0)	0	0	0	7 (10.9)	0
Myalgia	2 (4.3)	0	4 (26.7)	1 (6.7)	1 (50.0)	0	7 (10.9)	1 (1.6)
Vomiting	2 (4.3)	0	5 (33.3)	1 (6.7)	0	0	7 (10.9)	1 (1.6)

Abbreviation: TRAE, treatment-related adverse event.

immunosuppressants. Tumor necrosis factor alpha (TNF- α) inhibitors (such as infliximab) were permitted in steroid-sensitive settings to manage immune-mediated diarrhea/colitis, even in the nonrefractory low-grade setting.^{13,14} Oral or intravenous steroids and intravenous TNF- α inhibitors were prescribed per investigator's discretion. Steroids were typically prescribed at 0.5–1 mg/kg/day (including once, once daily, twice daily, 3 times/day, 4 times/day, every 6 hours, at bedtime, or as needed), while TNF- α inhibitors were typically prescribed at 5–10 mg/kg IV once. Patients who received TNF- α inhibitors were permitted to continue study drug if toxicity was resolved and grade ≤ 2 (Appendix Fig A1). If a patient had a grade 3 event related to BOT but unrelated to BAL, they were allowed to continue BAL. In total, there were 38 events of immune-mediated diarrhea/colitis, composed of the preferred terms of diarrhea, colitis, immune-mediated enteritis, autoimmune colitis, enterocolitis, and GI inflammation (Appendix Table A2), 37 (97.4%) of which were treated with immunosuppressants (TNF- α inhibitors and/or corticosteroids), and 33 (89.2%) of those resolved. Other common immune-mediated TRAEs included skin reactions, occurring in 17.2% of patients, and hypothyroidism in 10.9%. Nondiarrhea/colitis immune-mediated TRAEs were managed according to investigator's discretion.¹⁵

Efficacy

Patients evaluable for efficacy had a baseline and at least one postbaseline 6-week tumor imaging assessment for response (n = 52) with RECIST-confirmed investigator-assessed responses reported (Table 3). Overall, 10 of 52 evaluable patients achieved a response by RECIST 1.1 (ORR, 19.2% [95% CI, 9.6 to 32.5]). Responses were observed in patients with cAS, vAS, dLPS, soft tissue LMS, and UPS (Fig 1A, waterfall plot). There were no confirmed CRs, although there was one unconfirmed CR in a patient with cAS

(PR by RECIST). Importantly, responses were durable (Fig 1B, swimmer plot) with a median DOR of 21.7 months (95% CI, 1.9 to NR). Twenty-four additional patients achieved SD as best response, for a DCR of 65.4% (95% CI, 50.9 to 78.0). The CBR was 32.7% (95% CI, 20.3 to 47.1), and median PFS was 4.4 months (95% CI, 2.8 to 6.1), with a 6-month PFS rate of 36% (95% CI, 21.9 to 50.2; Fig 2A). With a median follow-up of 9.1 months (range, 1.4–39.7), median OS was NR (95% CI, 14.0 to NR), and the 12-month OS rate measured 69% (95% CI, 52.3 to 80.8; Fig 2B). Outside of RECIST, additional patients achieved clinical benefit with BOT/BAL including a patient with vAS who had pseudoprogression at 6 weeks (PD by RECIST) followed by a deep PR in target lesions, which persists off therapy to date (2 years). When assessed by iRECIST, the confirmed ORR was 21.2% in evaluable patients (95% CI, 11.1 to 34.7; Appendix Table A3).

Given the relatively large number of patients enrolled and evaluable for efficacy with AS (n = 18), we performed an exploratory analysis of efficacy in this subtype (Appendix Table A4 and Fig A2; Fig 3). The RECIST-confirmed ORR for AS was 27.8%, and 33.3% by iRECIST, with a DCR of 77.8% by RECIST 1.1, and 83.3% by iRECIST (Appendix Table A5). The CBR was 38.9% by RECIST 1.1 and 44.4% by iRECIST. The difference between RECIST 1.1 and iRECIST is related to the previously mentioned vAS patient with pseudoprogression (Appendix Fig A3). Importantly, RECIST-confirmed ORRs in patients with cAS and vAS were 22.2% and 33.3%, respectively (Appendix Table A6). The median PFS for patients with AS was 4.7 months (95% CI, 2.8 to NR), with a 6-month PFS rate of 48.4% (95% CI, 21.4 to 71.0), and a median DOR of 21.7 months (95% CI, 1.9 to NR).

Regarding additional subtypes, the ORR in soft tissue LMS was 13.3% (two PRs out of 15 patients), and an additional nine patients had clinical benefit, with a DCR of 73.3% (11/15).

TABLE 3. Summary of Efficacy Data in Efficacy-Evaluable Patients by RECIST 1.1 (n = 52)

Efficacy Measure	1 mg/kg Combo (n = 39)	2 mg/kg Combo (n = 11)	Crossover (n = 2)	Overall (n = 52)
ORR				
No. (%)	6 (15.4)	3 (27.3)	1 (50.0)	10 (19.2)
95% CI	(5.9 to 30.5)	(6.0 to 61.0)	(1.3 to 98.7)	(9.6 to 32.5)
BOR, No. (%)				
CR	0	0	0	0
PR	6 (15.4)	3 (27.3)	1 (50.0)	10 (19.2)
SD	20 (51.3)	4 (36.4)	0	24 (46.2)
PD	13 (33.3)	4 (36.4)	1 (50.0)	18 (34.6)
DCR				
No. (%)	24 (66.7)	7 (63.6)	1 (50.0)	34 (65.4)
95% CI	(49.8 to 80.9)	(30.8 to 89.1)	(1.3 to 98.7)	(50.9 to 78.0)
CBR				
No. (%)	12 (30.8)	4 (36.4)	1 (50.0)	17 (32.7)
95% CI	(17.0 to 47.6)	(10.9 to 69.2)	(1.3 to 98.7)	(20.3 to 47.1)
OS				
Median (95% CI)	NR (14.0 to NR)	6.4 (4.7 to NR)	NR (NR to NR)	NR (14.0 to NR)
12-month OS, % (95% CI)	73.3 (53.0 to 85.8)	47.1 (15.2 to 74.1)	100 (100 to 100)	69.0 (52.3 to 80.8)
PFS				
Median (95% CI)	4.4 (1.4 to 6.1)	4.7 (1.3 to NR)	NR (1.3 to NR)	4.4 (2.8 to 6.1)
6-month PFS, % (95% CI)	34.0 (18.2 to 50.6)	39.0 (10.6 to 67.3)	50.0 (6.0 to 91.0)	36.0 (21.9 to 50.2)
DOR				
Median (95% CI)	NR (3.4 to NR)	21.7 (1.9 to NR)	NR (NR to NR)	21.7 (1.9 to NR)

Abbreviations: BOR, best overall response; CBR, clinical benefit rate; CR, complete response; DCR, disease control rate; DOR, duration of response; NR, not reached; ORR, objective response rate; OS, overall survival; PD, progressive disease; PFS, progression-free survival; PR, partial response; SD, stable disease.

Furthermore, deep, durable responses were observed in two patients with dLPS, with an ORR of 50.0% (two of four patients) and DCR of 75.0%.

Correlative Studies

To investigate potential biomarkers of response to BOT/BAL, we examined peripheral IL-6 levels,¹⁶ PD-L1 expression, TMB, and FCGR3A status (Data Supplement, online only). Lower peripheral IL-6 levels were significantly associated with improved OS (Appendix Fig A4A). Patients were stratified on the basis of peripheral IL-6 levels, with a cutoff at the lower limit of quantification (LLOQ) of 0.63 pg/mL. Patients with IL-6 levels ≤LLOQ showed markedly improved OS compared with those with IL-6 >LLOQ ($P = .009$), with a hazard ratio (HR) of 0.15 (95% CI, 0.04 to 0.63).

Despite a trend toward improved survival in patients with PD-L1 CPS ≥1 compared with those with PD-L1 CPS <1, PD-L1 expression did not significantly correlate with OS (Appendix Fig A4B; $P = .26$; HR, 0.51 [95% CI, 0.15 to 1.65]). High TMB and FcγRIIIA (CD16)-V158F SNPs have been shown to be predictive markers of clinical response to ipilimumab in patients with advanced melanomas.¹⁷ However, in our cohort, patients with V/F or V/V genotypes had similar survival outcomes to those with the F/F genotype ($P = .45$; HR, 0.81

[95% CI, 0.24 to 1.94]; Appendix Fig A4C), consistent with the ability of BOT to bind strongly to high- and low-affinity CD16 receptors.⁸ Notably, most patients' tumors in our study exhibited low TMB (<10 mut/Mb), consistent with the low TMB observed among sarcomas compared with other cancer types, which may partially explain low response rates of sarcoma to conventional ICIs.^{18,19}

DISCUSSION

Most sarcomas are increasingly recognized as immunologically cold tumors. Apart from a few subtypes, transcriptomic data suggest that up to 80% of the most common subtypes of adult sarcomas have low TMB, defective antigen presentation, and low expression of innate and adaptive immune signatures.⁵ The two largest studies of combination CTLA-4/PD-1-directed therapies reported all-comer response rates of 16% for ipilimumab/nivolumab,²⁰ and 12% by immune-related RECIST (irRECIST) with tremelimumab/durvalumab.²¹ Patients with AS have derived greater benefit from ICIs, particularly cutaneous head and neck tumors with a UV signature that resembles melanoma.⁶ Angiosarcomas arising in visceral locations or primary breast sites more often exhibit mutations in *KDR*, *TP53*, and *PIK3CA*, which may explain decreased responses to ICIs.²²

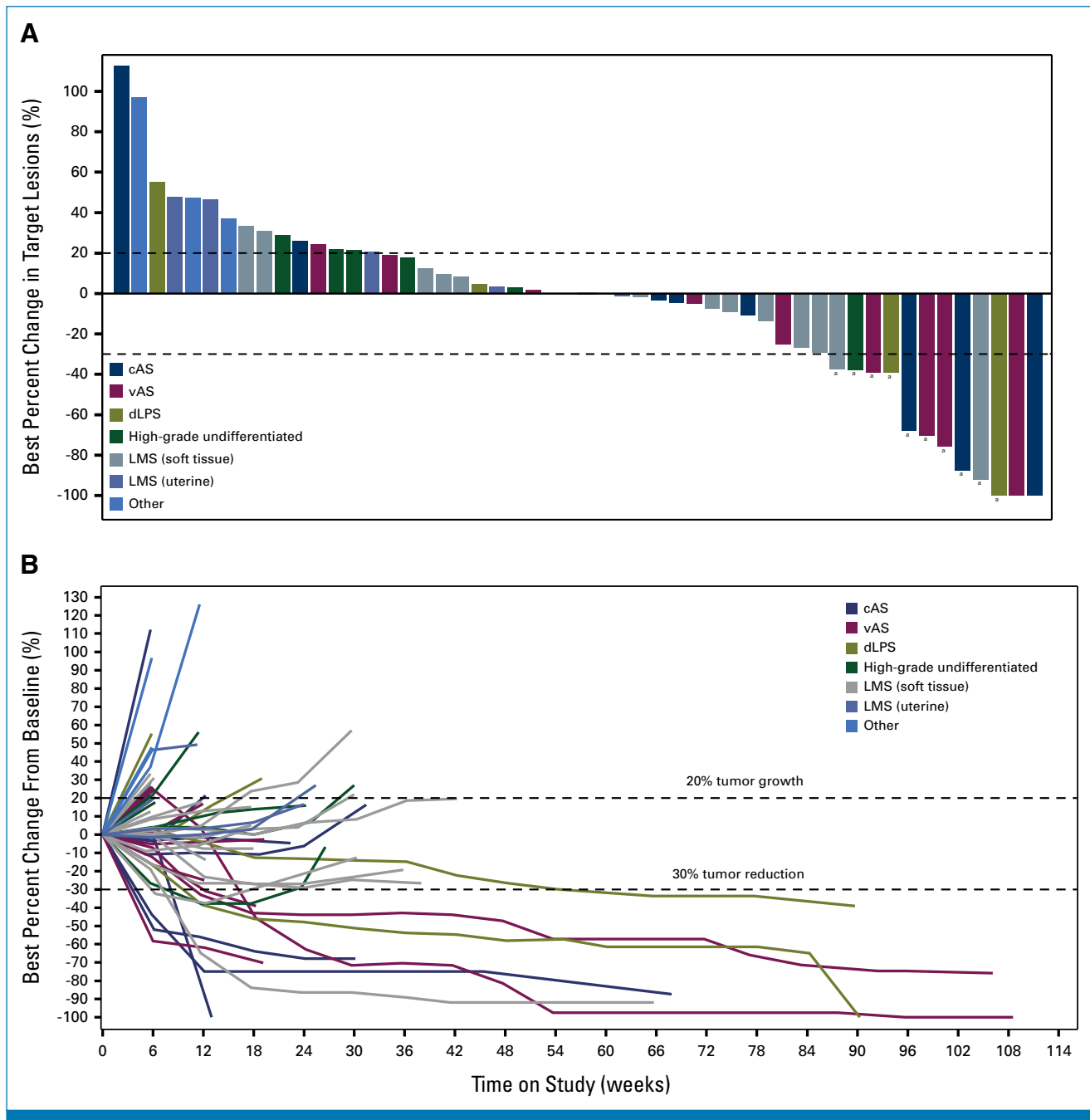


FIG 2. Clinical efficacy in efficacy-evaluable patients with sarcoma (n = 52). (A) Best overall response and (B) response over time. ^aA RECIST version 1.1-confirmed CR or PR. cAS, cutaneous angiosarcoma; CR, complete response; dLPS, dedifferentiated liposarcoma; LMS, leiomyosarcoma; PR, partial response; vAS, visceral angiosarcomas.

Antibodies targeted against CTLA-4 are thought to help improve engagement within the APC and T-cell synapse, helping to generate and expand tumor-specific T-cell priming, activation, and clonality, to be augmented by downstream PD-1 blockade. This is desirable in poorly immunogenic tumors such as sarcomas, and the response rate of the combination of ipilimumab/nivolumab performed better historically than nivolumab monotherapy.²⁰ Preclinical modeling and correlative analysis suggests BOT extends the reach of CTLA-4 blockade, with superior depletion of T_{reg}

cells, expansion of activated phenotypes of myeloid cells, along with potentiation of T-cell activation.⁸ Trial data suggests that BOT appears to improve efficacy of ICIs in immune-cold tumors, including MSS mCRC and ICI-refractory melanoma and non-small-cell lung cancer, among others.

In the current study, we observed clinically meaningful activity across a population of heavily pretreated sarcomas, with a RECIST-confirmed ORR of 19.2% and a

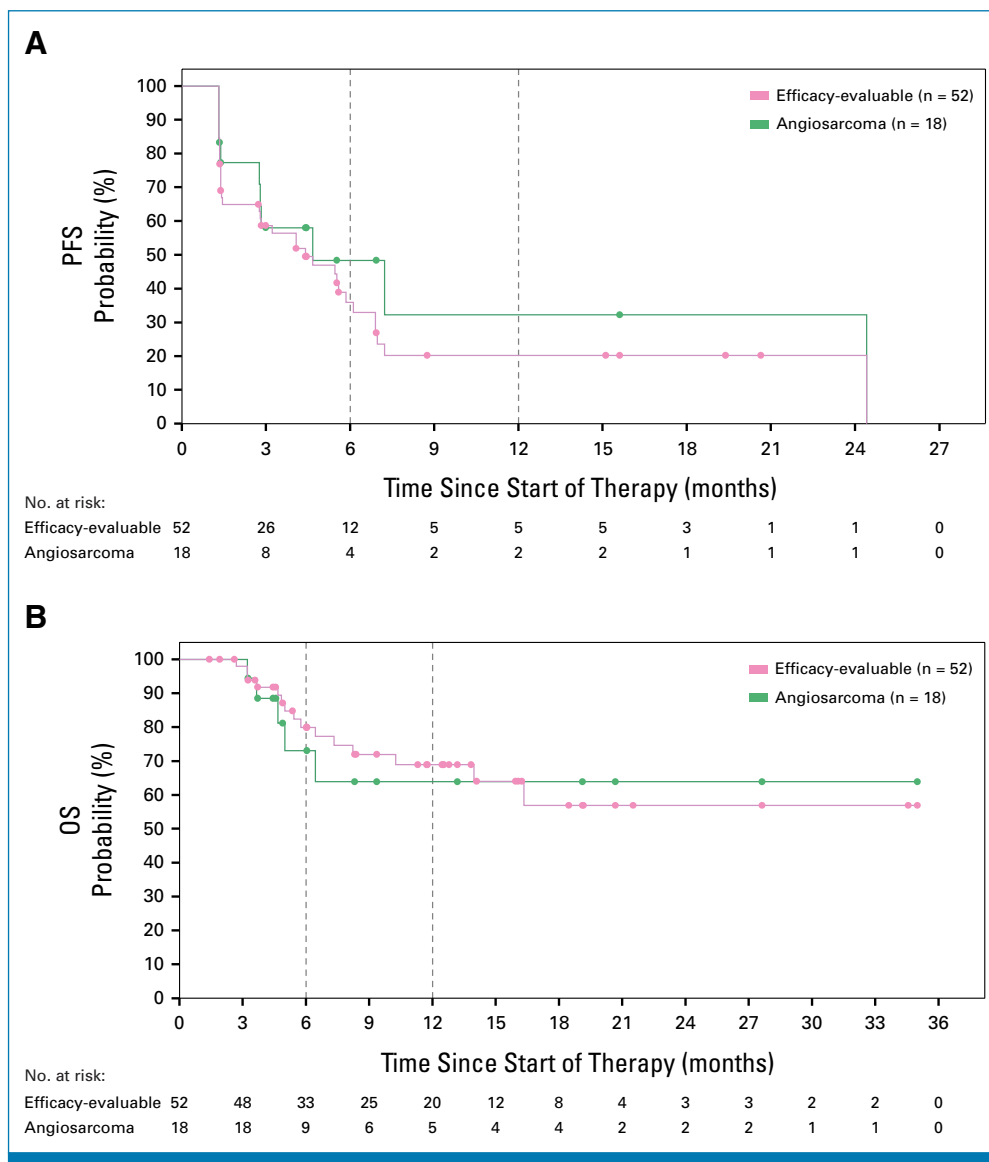


FIG 3. Clinical outcomes in all efficacy-evaluable patients with sarcoma (n = 52) and efficacy-evaluable patients with angiosarcoma (n = 18). (A) PFS and (B) OS. OS, overall survival; PFS, progression-free survival.

6-month PFS rate of 36.0%. Additionally, patients with AS had a RECIST-confirmed ORR of 27.8%. The ORRs for cAS and vAS subtypes measured 22.2% and 33.3%, respectively. These outcomes compare favorably to previous experience of ipilimumab/nivolumab, suggesting that BOT may offer superior activity in immune-cold sarcomas.

The overall safety profile was manageable and as expected for the class of ICIs. As with other tumor types included in the phase I trial, the most common treatment-related toxicity associated with BOT was diarrhea/colitis; however, most episodes could be managed successfully with early intervention including TNF- α inhibitors and rapid steroid tapering minimizing time without treatment. Importantly, this strategy represents an adapted paradigm for management of ICI-associated diarrhea/colitis, which should

improve safety profiles of other CTLA-4 inhibitors. The absence of other immune toxicities, such as myocarditis, pneumonitis, and hypophysitis, may be explained by the decreased complement binding resulting from Fc-engineering of BOT.

Our hypothesis-generating correlative data revealed complex relationships between biomarkers and treatment outcomes with BOT/BAL in sarcoma. We observed a trend toward improved survival in patients with PD-L1 CPS $\geq 1\%$ versus CPS $< 1\%$. Although PD-L1 has not been a conclusive biomarker for response to PD-1-directed therapy in sarcomas, this potential association with response to CTLA-4 blockade warrants further study.^{23,24} As expected, patients with sarcoma had low TMB (< 10 mut/Mb), further highlighting the potential of BOT/BAL to elicit responses in

low-TMB tumors. FCGR3A SNP status, previously shown to affect response to ipilimumab in melanoma, did not significantly affect OS, consistent with the ability of BOT to bind both high- and low-affinity receptors, and overcome the limitations associated with FCGR3A polymorphisms observed with conventional IgG1 anti-CTLA-4 antibodies. Finally, lower peripheral IL-6 levels were significantly associated with improved OS, consistent with observations in other cancers.¹⁶

An emerging but controversial biomarker being explored in sarcomas is the presence of tertiary lymphoid structures (TLS), with TLS presence associated with pembrolizumab efficacy in advanced soft tissue sarcomas.^{5,25-27} Although our study did not assess TLS, our previous data describing mechanisms of action of BOT in patients with advanced solid cancers suggest that enhanced Fc-effector functions could promote conditions favorable for TLS formation through multiple pathways including activation of type-1 conventional dendritic cells, expansion of B and activated CD4⁺ T-follicular-helper cells, and induction of key

chemokines associated with immune recruitment to the tumor microenvironment.⁸

Our study has several limitations. First, our efficacy results are reported for patients who had at least one postbaseline 6-week imaging scan. As is common in phase I studies, there were 12 patients who enrolled who did not reach protocol-defined imaging assessments. Reasons for discontinuation included rapid progression by RECIST 1.1, disease-related adverse events leading to significant medical complications, or death. Similarly, PFS and OS curves are heavily censored because of several patients withdrawing consent early. Importantly, immunotherapy requires time to exert its effects.

Overall, we observed promising efficacy and safety in a relatively large cohort of patients with heavily pretreated and heterogeneous sarcomas, including exciting activity in vAS and disease control in LMS. Further exploration of activity and safety in selected sarcoma subsets, including ASs, is warranted.

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Botensilimab (Fc-enhanced anti–cytotoxic lymphocyte-association protein-4 antibody) Plus Balstilimab (anti–PD-1 antibody) in Patients With Relapsed/Refractory Metastatic Sarcomas

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APPENDIX

TABLE A1. Overall Safety Summary in All Treated Patients With Sarcoma (n = 64)

Adverse Event	1 mg/kg Combo (n = 47), No. (%)	2 mg/kg Combo (n = 15), No. (%)	Crossover (n = 2), No. (%)	Overall (n = 64), No. (%)
Any TEAE	47 (100.0)	15 (100.0)	2 (100.0)	64 (100.0)
Grade ≥ 3 TEAE	29 (61.7)	10 (66.7)	1 (50.0)	40 (62.5)
Serious TEAE	36 (76.6)	9 (60.0)	1 (50.0)	46 (71.9)
TEAE leading to death	4 (8.5)	2 (13.3)	0	6 (9.4)
TEAE leading to treatment interruption	28 (59.6)	7 (46.7)	1 (50.0)	36 (56.3)
TEAE leading to treatment discontinuation	15 (31.9)	1 (6.7)	1 (50.0)	17 (26.6)
Any TRAE	37 (78.7)	14 (93.3)	2 (100.0)	53 (82.8)
Grade ≥ 3 TRAE	6 (12.8)	5 (33.3)	0	11 (17.2)
Serious TRAE	11 (23.4)	6 (40.0)	0	17 (26.6)
TRAE leading to death	0	0	0	0
TRAE leading to treatment interruption	18 (38.3)	4 (26.7)	1 (50.0)	23 (35.9)
TRAE leading to treatment discontinuation	8 (17.0)	0	0	8 (12.5)
BOT-related TRAE leading to BOT discontinuation	8 (17.0)	0	0	8 (12.5)

Abbreviations: BOT, botensilimab; TEAE, treatment-emergent adverse event; TRAE, treatment-related adverse event.

TABLE A2. All Immune-Mediated Treatment-Related Adverse Events in All Treated Patients With Sarcoma (n = 64)

Immune-Mediated TRAE	Overall (n = 64), No. (%)	
	Any Grade	Grade 3
Any	33 (51.6)	8 (12.5)
Diarrhea/colitis ^a	23 (35.9)	5 (7.8)
Skin adverse reactions ^b	11 (17.2)	0
Hypothyroidism ^c	7 (10.9)	0
Other reactions ^d	7 (10.9)	1 (1.6)
Hyperthyroidism ^e	4 (6.3)	0
Hepatitis ^f	3 (4.7)	1 (1.6)
Myositis/rhabdomyolysis ^g	1 (1.6)	0
Nervous system disorder ^h	1 (1.6)	1 (1.6)
Ocular disorder ⁱ	1 (1.6)	1 (1.6)
Pancreatitis ^j	1 (1.6)	0
Thyroiditis ^k	1 (1.6)	0

Abbreviation: TRAE, treatment-related adverse event.

^aComprised the preferred terms of diarrhea, colitis, immune-mediated enterocolitis, autoimmune colitis, enterocolitis, and GI inflammation.

^bComprised the preferred terms of rash maculopapular, rash, immune-mediated dermatitis, and lichen sclerosus.

^cComprised the preferred term hypothyroidism.

^dComprised the preferred terms of nausea, vomiting, polyarthritis, and synovitis.

^eComprised the preferred term hyperthyroidism.

^fComprised the preferred terms ALT increased, AST increased, blood alkaline phosphatase increased, and hypertransaminasemia.

^gComprised the preferred term immune-mediated myositis.

^hComprised the preferred term aseptic meningitis.

ⁱComprised the preferred term scleritis.

^jComprised the preferred term pancreatitis.

^kComprised the preferred term blood thyroid-stimulating hormone increased.

TABLE A3. Summary of Efficacy Data in Efficacy-Evaluable Patients With Sarcoma by iRECIST (n = 52)

Efficacy Measure	1 mg/kg Combo (n = 39)	2 mg/kg Combo (n = 11)	Crossover (n = 2)	Overall (n = 52)
ORR				
No. (%)	7 (17.9)	3 (27.3)	1 (50.0)	11 (21.2)
95% CI	(7.5 to 33.5)	(6.0 to 61.0)	(1.3 to 98.7)	(11.1 to 34.7)
BOR, no. (%)				
CR	0	0	0	0
PR	7 (18.0)	3 (27.3)	1 (50.0)	11 (21.2)
SD	20 (51.3)	4 (36.4)	0	24 (46.2)
PD	12 (30.8)	4 (36.4)	1 (50.0)	17 (32.7)
DCR				
No. (%)	27 (69.2)	7 (63.6)	1 (50.0)	35 (67.3)
95% CI	(52.4 to 83.0)	(30.8 to 89.1)	(1.3 to 98.7)	(52.9 to 79.7)
CBR				
No. (%)	13 (33.3)	4 (36.4)	1 (50.0)	18 (34.6)
95% CI	(19.1 to 50.2)	(10.9 to 69.2)	(1.3 to 98.7)	(22.0 to 49.1)
OS				
Median (95% CI)	NR (14.0 to NR)	6.4 (4.7 to NR)	NR (NR to NR)	NR (14.0 to NR)
12-month OS, % (95% CI)	73.3 (53.0 to 85.8)	47.1 (15.2 to 74.1)	100 (100 to 100)	69 (52.3 to 80.8)
PFS				
Median (95% CI)	4.4 (1.4 to 6.1)	4.7 (1.3 to NR)	NR (1.3 to NR)	4.4 (2.8 to 6.1)
6-month PFS, % (95% CI)	34.0 (18.2 to 50.6)	39.0 (10.6 to 67.3)	50.0 (6.0 to 91.0)	36.0 (21.9 to 50.2)
DOR				
Median (95% CI)	NR (3.4 to NR)	21.7 (1.9 to NR)	NR (NR to NR)	21.7 (3.4 to NR)

Abbreviations: BOR, best overall response; CBR, clinical benefit rate; CR, complete response; DCR, disease control rate; DOR, duration of response; iRECIST, modified RECIST for immune-based therapeutics; NR, not reached; ORR, objective response rate; OS, overall survival; PD, progressive disease; PFS, progression-free survival; PR, partial response; SD, stable disease.

TABLE A4. Summary of Efficacy Data in Efficacy-Evaluable Patients With Angiosarcoma by RECIST 1.1 (n = 18)

Efficacy Measure	1 mg/kg Combo (n = 8)	2 mg/kg Combo (n = 10)	Overall (n = 18)
ORR			
No. (%)	2 (25.0)	3 (30.0)	5 (27.8)
95% CI	(3.2 to 65.1)	(6.7 to 65.2)	(9.7 to 53.5)
BOR, No. (%)			
CR	0	0	0
PR	2 (25.0)	3 (30.0)	5 (27.8)
SD	5 (62.5)	4 (40.0)	9 (50.0)
PD	1 (12.5)	3 (30.0)	4 (22.2)
DCR			
No. (%)	7 (87.5)	7 (70.0)	14 (77.8)
95% CI	(47.3 to 99.7)	(34.8 to 93.3)	(52.4 to 93.6)
CBR			
No. (%)	3 (37.5)	4 (40.0)	7 (38.9)
95% CI	(8.5 to 75.5)	(12.2 to 73.8)	(17.3 to 64.3)
OS			
Median (95% CI)	NR (3.6 to NR)	NR (3.2 to NR)	NR (5.0 to NR)
12-month OS, % (95% CI)	85.7 (33.4 to 97.9)	52.5 (16.8 to 79.3)	63.9 (32.6 to 83.6)
PFS			
Median (95% CI)	NR (1.3 to NR)	4.7 (1.3 to NR)	4.7 (2.8 to NR)
6-month PFS, % (95% CI)	58.3 (18.0 to 84.4)	42.9 (11.4 to 71.8)	48.4 (21.4 to 71.0)
DOR			
Median (95% CI)	NR (3.4 to NR)	21.7 (1.9 to NR)	21.7 (1.9 to NR)

Abbreviations: BOR, best overall response; CBR, clinical benefit rate; CR, complete response; DCR, disease control rate; DOR, duration of response; NR, not reached; ORR, objective response rate; OS, overall survival; PD, progressive disease; PFS, progression-free survival; PR, partial response; SD, stable disease.

TABLE A5. Summary of Efficacy Data in Efficacy-Evaluable Patients With Angiosarcoma by iRECIST (n = 18)

Efficacy Measure	1 mg/kg Combo (n = 8)	2 mg/kg Combo (n = 10)	Overall (n = 18)
ORR			
No. (%)	3 (37.5)	3 (30.0)	6 (33.3)
95% CI	(8.5 to 75.5)	(6.7 to 65.2)	(13.3 to 59.0)
BOR, No. (%)			
CR	0	0	0
PR	3 (37.5)	3 (30.0)	6 (33.3)
SD	5 (62.5)	4 (40.0)	9 (50.0)
PD	0	3 (30.0)	3 (16.7)
DCR			
No. (%)	8 (100.0)	7 (70.0)	15 (83.3)
95% CI	(63.1 to 100.0)	(34.8 to 93.3)	(58.6 to 96.4)
CBR			
No. (%)	4 (50.0)	4 (40.0)	8 (44.4)
95% CI	(15.7 to 84.3)	(12.2 to 73.8)	(21.5 to 69.2)
OS			
Median (95% CI)	NR (3.6 to NR)	NR (3.2 to NR)	NR (5.0 to NR)
12-month OS, % (95% CI)	85.7 (33.4 to 97.9)	52.5 (16.8 to 79.3)	63.9 (32.6 to 83.6)
PFS			
Median (95% CI)	NR (1.3 to NR)	4.7 (1.3 to NR)	4.7 (2.8 to NR)
6-month PFS, % (95% CI)	58.3 (18.0 to 84.4)	42.9 (11.4 to 71.8)	48.4 (21.4 to 71.0)
DOR			
Median (95% CI)	NR (NR to NR)	21.7 (1.9 to NR)	21.7 (1.9 to NR)

Abbreviations: BOR, best overall response; CBR, clinical benefit rate; CR, complete response; DCR, disease control rate; DOR, duration of response; iRECIST, modified RECIST for immune-based therapeutics; NR, not reached; ORR, objective response rate; OS, overall survival; PD, progressive disease; PFS, progression-free survival; PR, partial response; SD, stable disease.

TABLE A6. Summary of Efficacy Data in Efficacy-Evaluable Patients With Cutaneous (n = 9) and Visceral (n = 9) Angiosarcoma by RECIST 1.1

Efficacy Measure	Cutaneous Angiosarcoma			Visceral Angiosarcoma		
	1 mg/kg Combo (n = 2)	2 mg/kg Combo (n = 7)	Overall (n = 9)	1 mg/kg Combo (n = 6)	2 mg/kg Combo (n = 3)	Overall (n = 9)
ORR						
No. (%)	1 (50.0)	1 (14.3)	2 (22.2)	1 (16.7)	2 (66.7)	3 (33.3)
95% CI	(1.3 to 98.7)	(0.4 to 57.9)	(2.8 to 60.0)	(0.4 to 64.1)	(9.4 to 99.2)	(7.5 to 70.1)
BOR, No. (%)						
CR	0	0	0	0	0	0
PR	1 (50.0)	1 (14.3)	2 (22.2)	1 (16.7)	2 (66.7)	3 (33.3)
SD	1 (50.0)	3 (42.9)	4 (44.4)	4 (66.7)	1 (33.3)	5 (55.6)
PD	0	3 (42.9)	3 (33.3)	1 (16.7)	0	1 (11.1)
DCR						
No. (%)	2 (100.0)	4 (57.1)	6 (66.7)	5 (83.3)	3 (100.0)	8 (88.9)
95% CI	(15.8 to 100.0)	(18.4 to 90.1)	(29.9 to 92.5)	(35.9 to 99.6)	(29.2 to 100.0)	(51.8 to 99.7)
CBR						
No. (%)	2 (100.0)	2 (28.6)	4 (44.4)	1 (16.7)	2 (66.7)	3 (33.3)
95% CI	(15.8 to 100.0)	(3.7 to 71.0)	(13.7 to 78.8)	(0.4 to 64.1)	(9.4 to 99.2)	(7.5 to 70.1)
DOR						
Median (95% CI)	NR (NR to NR)	NR (NR to NR)	NR (NR to NR)	NR (NR to NR)	11.8 (1.9 to NR)	21.7 (1.9 to NR)
Follow-up, months						
Median (range)	7.2 (6.0 to 8.3)	6.4 (3.2 to 20.7)	6.4 (3.2 to 20.7)	8.6 (3.6 to 27.6)	4.7 (1.4 to 39.7)	7.4 (1.4 to 39.7)

Abbreviations: BOR, best overall response; CBR, clinical benefit rate; CR, complete response; DCR, disease control rate; DOR, duration of response; NR, not reached; ORR, objective response rate; OS, overall survival; PD, progressive disease; PFS, progression-free survival; PR, partial response; SD, stable disease.

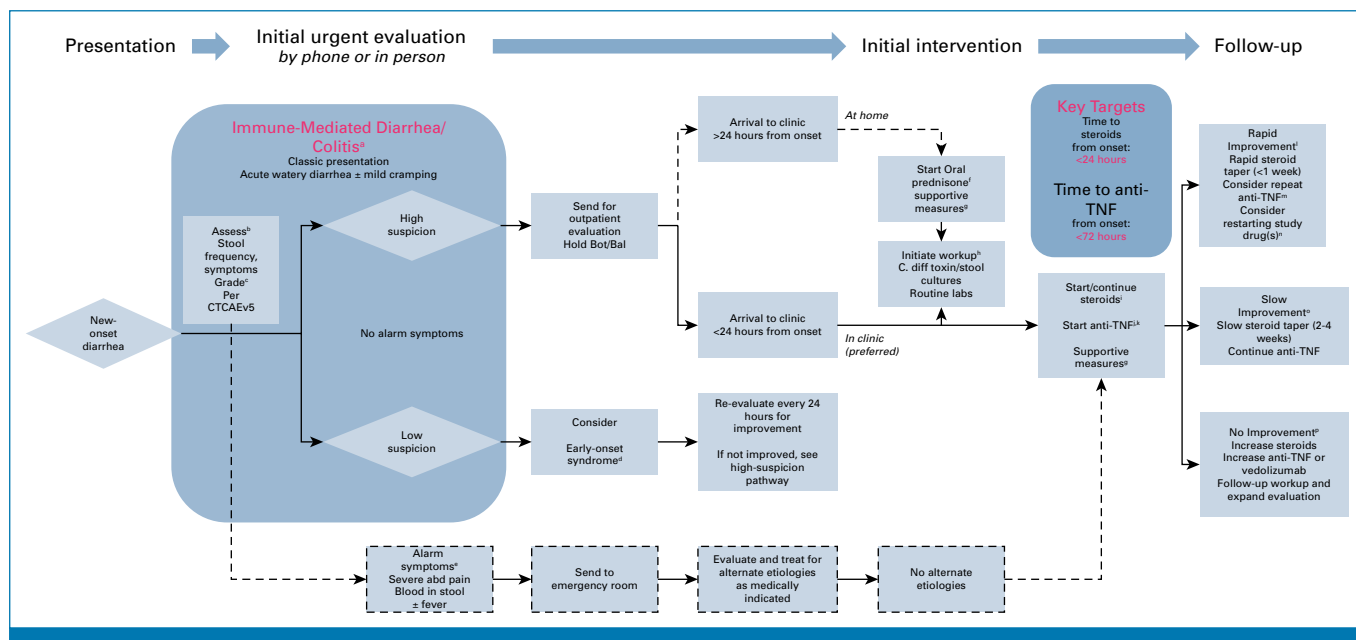


FIG A1. Algorithm for management of immune-mediated diarrhea/colitis. This algorithm reflects expert guidance and evolving literature on the management of immune-mediated colitis. It is intended to be used for patients receiving BOT-based therapy. This algorithm is not intended to replace the independent medical or professional judgment of physicians or other health care providers in the context of individual clinical circumstances to determine a patient's care. ^aAlthough the CTCAE version 5.0 classifies and grades diarrhea and colitis as separate entities, in the clinical setting of ICI therapy for cancer, immune-related diarrhea and colitis occur along a common spectrum. Because there is an underlying common pathology, and inflammation often involves both the large and small intestines, immune-related diarrhea and colitis are considered and discussed jointly.²⁸ ^bAssess patient as soon as possible with a goal of intervening within 24 hours of the onset of symptoms. The initial assessment may be over the phone or in person, but full workup requires in-person evaluation as soon as possible. Assessment should include number of stools above baseline, consistency (ie, watery), recent illnesses, infectious exposures, and recent antibiotics. Also assess for potential alarm symptoms (see 5 below). ^cCTCAEv5 grading: grade 1: increase of less than four stools per day over baseline; or mild increase in ostomy output over baseline. Grade 2: increase of four to six stools per day over baseline; or moderate increase in ostomy output over baseline; limiting instrumental ADLs. Grade 3: increase of ≥seven stools per day over baseline; or severe increase in ostomy output over baseline; limiting self-care ADLs; hospitalization indicated. Grade 4: life-threatening consequences, urgent intervention indicated. Grade 5: death. ^dIn a minority of patients treated with BOT, there is an early-onset, self-limited syndrome that may occur anywhere from hours to 1-2 weeks after infusion. This syndrome may include any of the following symptoms: fatigue, fever/chills, malaise, nausea, vomiting, or diarrhea. This syndrome does not represent immune-related diarrhea/colitis as it is generally self-limited and can be treated symptomatically with nonsteroidal anti-inflammatory drugs (consider 500 mg naproxen twice daily with appropriate GI prophylaxis if tolerated)/acetaminophen/±antimotility agents and supportive measures. Recommend careful monitoring of any patient who appears to have this syndrome to ensure resolution. The early-onset syndrome may reoccur with subsequent doses of BOT. However, if it did not occur with the first dose, it is unlikely to occur with subsequent doses and alternate etiologies should be considered, especially immune-related adverse events such as diarrhea/colitis. Timing of symptom onset in relation to BOT treatment is important. If diarrhea/nausea/vomiting occur >1-2 weeks after dosing, suspicion should be higher for immune-related diarrhea/colitis/enteritis and lower for the early-onset syndrome, whereas if symptoms develop within <1 week, immune-related diarrhea/colitis requiring steroids is extremely unlikely. ^eAny alarm symptom associated with diarrhea/colitis including but not limited to high-grade fever, severe abdominal pain, and blood in stool (hematochezia or melena) or hemodynamic instability warrant urgent evaluation. Note that fever may be seen with early-onset syndrome; however, this is a diagnosis of exclusion, especially if symptom onset is >1 week after dosing. ^fIf patient is unable to present for in-person evaluation in <24 hours, recommend utilization of oral prednisone that was provided at C1D1 (patient education and prescription) until they can present to medical facility for workup and treatment. For grade 1-2: prednisone 0.5-1 mg/kg/d. For grade 3-4: prednisone 1 mg/kg/d. ^gRecommend supportive measures immediately: BRAT diet and sufficient oral fluid intake, especially electrolyte-rich fluids. Ensure that any stool softeners and/or laxatives are discontinued. Antimotility agents are ineffective in the setting of immune-mediated diarrhea/colitis. ^hRecommend medical workup for acute-onset diarrhea of infectious or noninfectious etiology, which includes but is not limited to C. diff toxin, stool cultures, and ova/parasite examination, complete blood count, GI viral panel, complete metabolic panel, amylase/lipase, and inflammatory markers per provider discretion. C. diff is the most common alternate diagnosis and may co-occur with immune-related diarrhea/colitis. ⁱIf not already initiated, begin steroids. For grade 1-2: prednisone 0.5-1 mg/kg/d or equivalent. For grade 3-4: prednisone 1 mg/kg/d or equivalent, intravenous administration preferred if this would not result in a delay. ^jOnce patient has been assessed and nonimmune causes are effectively ruled out, we recommend treatment with infliximab (5 mg/kg) or equivalent, regardless of response to steroids or grade of diarrhea. Consider 10 mg/kg infliximab or equivalent for grade 4, or if severe hypoalbuminemia (albumin <2.5) is present. ^kIf tuberculosis testing (ie, Quantiferon Gold) is positive, then infliximab should not be given unless deemed acceptable by the treating provider on the basis of further workup and intervention, if the risk of anti-TNF is considered unacceptable, consider vedolizumab as an alternative. ^lRapid improvement of symptoms to grade 1 or less within 48 hours of treatment. If infliximab is administered and symptoms improve to ≤grade 1, a rapid steroid taper of <1 week is recommended. ^mIf steroid taper is rapid and in discussion with the sponsor, the plan is for the patient to resume BOT and administer three (continued on following page)

FIG A1. (Continued). doses of infliximab in total according to the prescribing information (at 0, 2, and 6 weeks) regardless of symptom resolution. If steroid taper is rapid but there is no plan to resume BOT, then additional dosing of infliximab is based on clinical judgment. ^aOnly patients who meet any of the following criteria will be considered eligible to restart BOT: (1) Diarrhea was not immune-mediated, or (2) did not exceed grade 2, responded rapidly to treatment, and received at least one dose of anti-TNF and will receive subsequent doses anti-TNF (total of at least three doses at 0, 2 and 6 weeks). Patients can restart BAL after discussion with sponsor even if diarrhea/colitis was grade 3 or higher, provided it was associated with BOT and not BAL monotherapy. ^oIf symptom duration requires a longer steroid taper, administer three doses infliximab total according to the prescribing information (at 0, 2, and 6 weeks). ^pIf symptoms do not improve within 48 hours after infliximab treatment, recommend re-evaluation for nonimmune causes of diarrhea and/or GI consultation for consideration of endoscopy. If immune-mediated etiology is still suspected, increase the dose of corticosteroids to a dose of 1-2 mg/kg/d prednisone or equivalent (ideally IV administered) and consider redosing infliximab at a higher dose (eg, 10 mg/kg). Consider treatment with vedolizumab (may require dosing by level if hypoalbuminemia present) or additional immunosuppressive therapy. abd, abdominal; ADL, activities of daily living; BOT/BAL, botensilimab/balstilimab; BRAT, bananas, rice, apple sauce, and toast; C. diff, clostridium difficile; CTCAEv5, Common Terminology Criteria for Adverse Events version 5; ICI, immune checkpoint inhibitor; IV, intravenously; TNF, tumor necrosis factor.

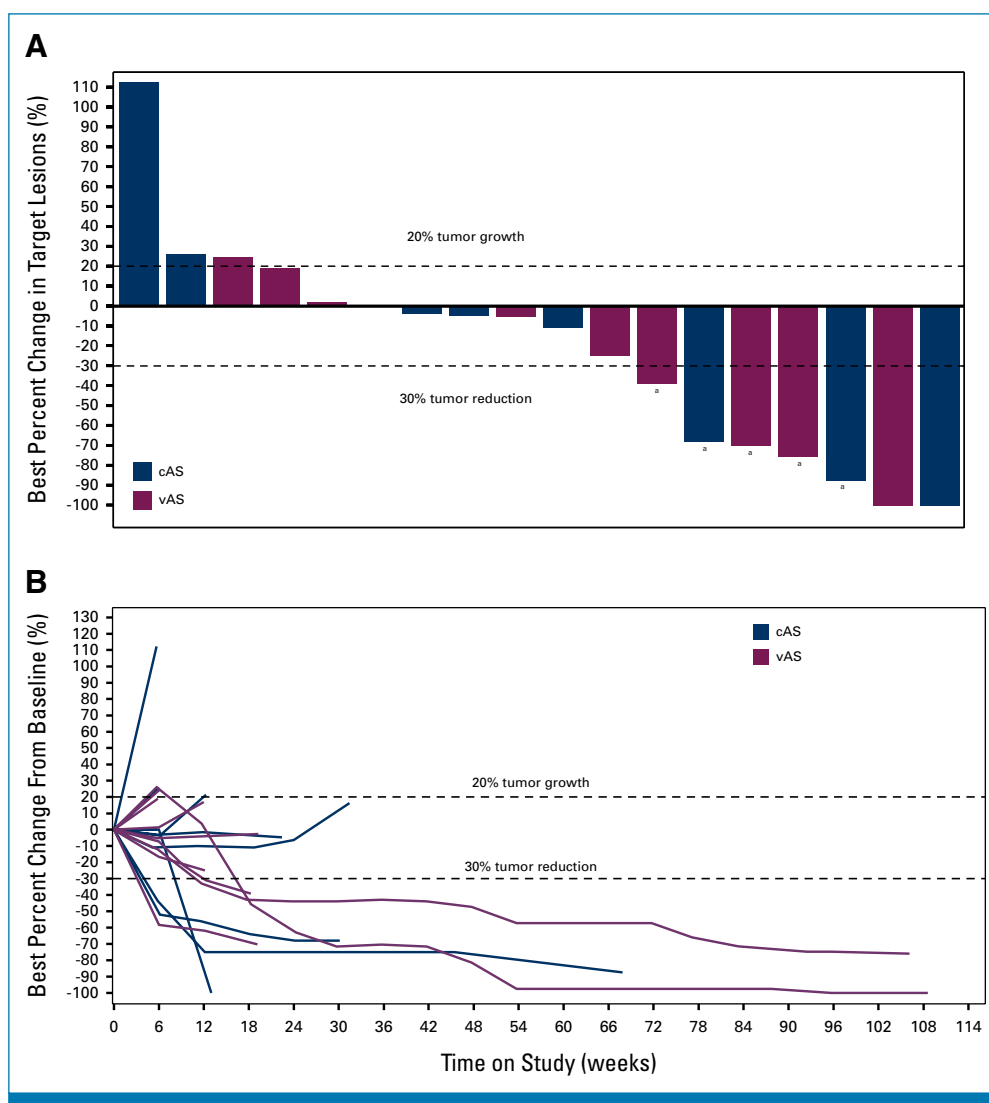


FIG A2. Clinical efficacy in efficacy evaluable patients with angiosarcoma (n = 18). (A) Best overall response; (B) response over time. ^aA RECIST version 1.1-confirmed CR or PR. cAS, cutaneous angiosarcoma; CR, complete response; PR, partial response; vAS, visceral angiosarcomas.

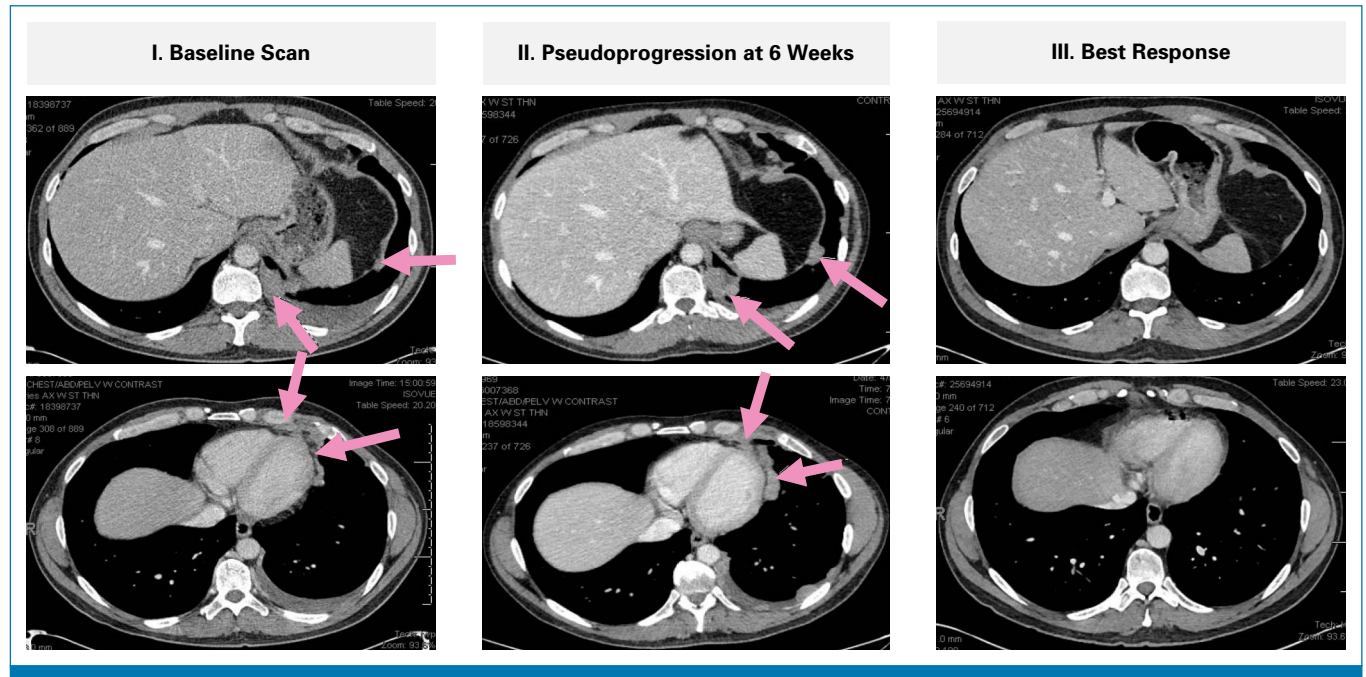


FIG A3. Time course showcasing an iRECIST responder. Imaging scans from a patient with visceral angiosarcoma who had pseudoprogression at 6 weeks (progressive disease by RECIST) followed by a deep partial response in a target lesion that has been maintained for 3 years and is now off therapy (for 2 years). iRECIST, modified RECIST for immune-based therapeutics.

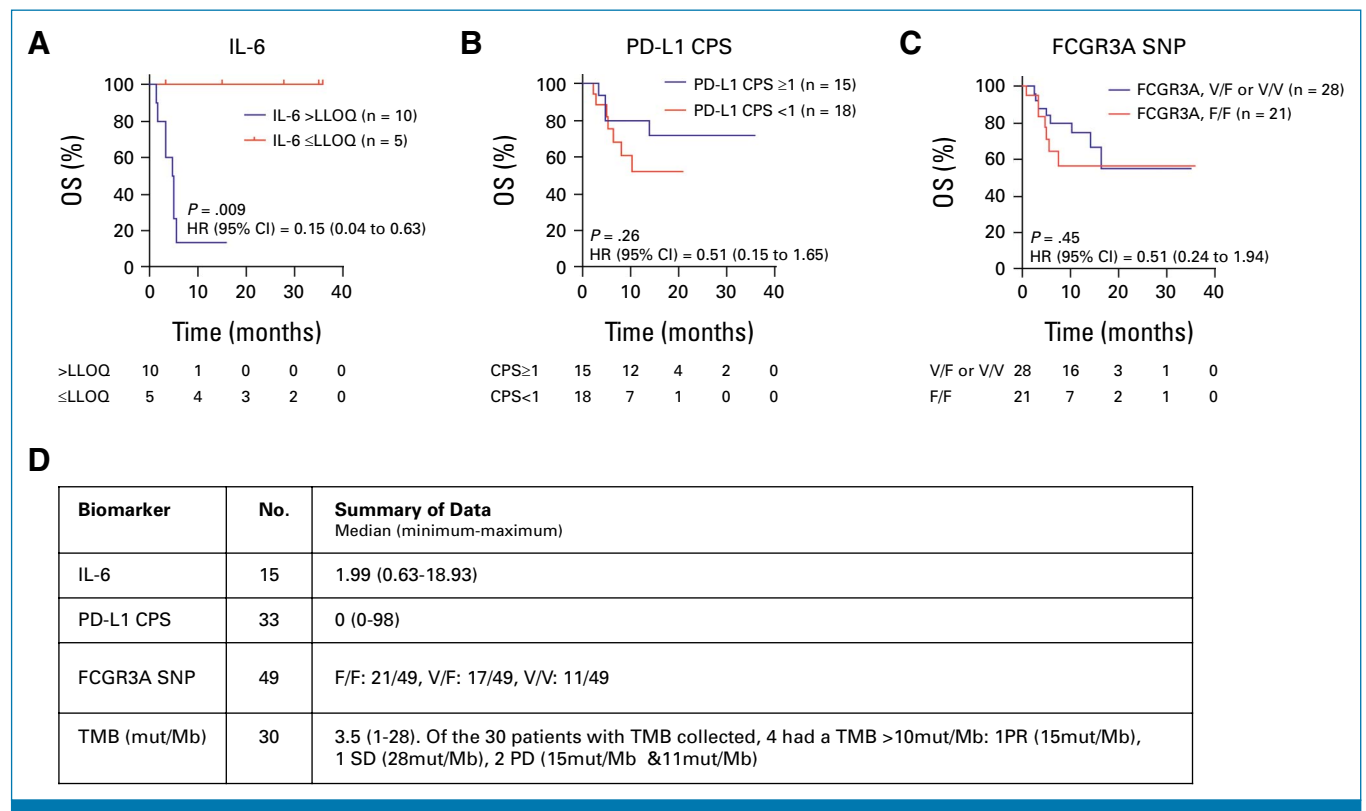


FIG A4. Evaluation of potential biomarkers of response to BOT and BAL treatment in patients with sarcoma. Comparison of OS of patients segregated by (A) peripheral IL-6 levels above or below the assay LLoQ; (B) combined positive score of PD-L1 expression above or below 1; and (C) germline FCGR3A p.V158F polymorphism. Data analyzed with log-rank (Mantel-Cox) test for *P* values and Mantel-Haenszel HR. (D) Summary data table of peripheral IL-6 levels 20, PD-L1 expression, TMB, and FCGR3A polymorphism (SNP) status. BAL, balstilimab; BOT, botensilimab; CPS, combined proportional score; HR, hazard ratio; IL, interleukin; LLOQ, lower limit of quantification; mut/Mb, mutations/megabase; OS, overall survival; PD, progressive disease; PR, partial response; SD, stable disease; SNP, single-nucleotide polymorphism; TMB, tumor mutational burden.